

5 Experimental Setup

5.1 Data and Evaluation Populations

We evaluate WANDS [1] and the US English portion of the small-version ESCI test split [2]. WANDS retains its full catalog of 42,994 products. We use the existing split of 96 development and 384 held-out queries; the 308 held-out queries with at least one Exact product contribute 21,299 query-product pairs. ESCI uses the existing deterministic sample of 500 queries and the union of their judged products, yielding a fixed catalog of 10,076 products. Its 499 queries with at least one E-labeled product contribute 4,434 pairs. Every primary strategy comparison within a dataset retains the same catalog and eligible query population; fully-fitting diagnostics restrict targets only. These are evaluations with incomplete relevance judgments, not estimates of access to every truly relevant product.

WANDS attributes are the complete pipe-separated entries in its product-features field. ESCI attributes are the nonempty brand, color, and complete bullet-point fields; the bullet-point field is retained as one entry. ESCI therefore has at most three entries per product. Titles, descriptions, and other non-attribute fields retain their text and section placement. Duplicate attribute occurrences are preserved. Data preparation removes duplicate judgments and excludes contradictory WANDS query-product labels.

5.2 Encoders and Ordering Schedules

The sensitivity and pure-canonicalization experiments use frozen all-MiniLM-L6-v2, bge-base-en-v1.5, and gte-modernbert-base, abbreviated MiniLM, BGE, and GTE. Their input limits are 256, 512, and 8,192 tokens, respectively; MiniLM uses attention-mask mean pooling, and BGE and GTE use CLS pooling. We retain empty query prefixes and unit-normalized embeddings. Retrieval uses exact FP32 dot-product scores and resolves ties by ascending fixed catalog index. Pinned revisions and execution details accompany the artifacts.

The seven schedules are source order (C0), reverse attribute order (C1), and five product-dependent shuffles (C2s1–C2s5), using seeds 20260911–20260915. Both interventions follow Section 3. Fully-fitting eligibility requires every complete input, including special tokens, to fit the corresponding encoder limit under all seven schedules; only evaluated targets are filtered, while all competitors remain.

The three canonical rules are fixed and all are reported. Aggregation and hybrid comparisons cover BGE and MiniLM on both datasets. Full-record aggregation uses $M = 2$ for this common comparison; the existing BGE sweep additionally covers $M = 4, 7$. Its view seeds, 20260971–20260977, differ from the evaluation-schedule seeds. Set-Mean uses equal weights, BM25 uses $k_1 = 1.2$ and $b = 0.75$, and hybrid retrieval uses RRF with constant 60.

5.3 Evaluation and Comparison Protocol

We report query-macro Recall, persistent inclusion, VI, and persistent omission at $K = 20, 100$, with pair-micro results kept separate. Membership comparisons use the same H_q and distinguish gains, losses, and their net difference. Exact within-query cancellation requires equal positive integer gain and loss counts. Source-order

Recall and the raw seven-schedule mean are separate references; target-only inclusion is not labeled catalog-wide Recall.

All common strategy, order, canonicalization, and cutoff analyses use 10,000 paired query-cluster bootstrap draws with seed 2026091701 and descriptive 95% percentile intervals. Each draw retains a query’s products and compared conditions together. Intervals are uncorrected for multiple comparisons; containing zero does not establish equivalence. The common strategy table reports seven-schedule mean Recall and VI. Paired intervals, source-order comparisons, membership counts, and persistent states are supplied separately. Hybrid uses the complete 1-based dense and BM25 ranks for each schedule; its C0 results exactly reproduce the saved source-order hybrid. Historical embeddings are reused without re-encoding, and reconstructed highest-label raw ranks agree with the authoritative saved ranks in all 28 hybrid conditions. Historical strategy selection used WANDS development queries, whereas the three pure canonical rules were fixed before their additional retrieval runs. Earlier outcomes had already been inspected; these analyses are not a newly untouched confirmation set, and no test-set winner is selected. Cost distinguishes historical offline cache construction, query encoding, exact scoring/sorting, and dense/sparse array bytes. Cache-hit elapsed times are not cold-build costs; local exact retrieval timings are not ANN deployment measurements. Additional diagnostics use $K=20,50,100,200,500,1000$ on fixed target support within each curve.

6 Results

We first examine candidate inclusion under attribute permutations, then quantify membership changes relative to Recall, and finally compare representation and retrieval controls. All main estimates use the primary query populations in Section 5.1.

6.1 RQ1: Attribute Order and Relevant-Product Inclusion

Catalog-wide permutations change relevant-product inclusion in every dataset-encoder setting. Across the seven schedules, VI@20 is 19.58%, 11.93%, and 14.47% on WANDS for MiniLM, BGE, and GTE, respectively; the corresponding ESCI values are 4.04%, 3.66%, and 6.91%. These proportions measure crossing within the tested family, rather than the probability of omission after one arbitrary update.

The effect persists when only a target’s representation changes. With every competitor fixed at its source-order representation, target-only VI@20 is 19.45%, 10.83%, and 14.55% on WANDS, and 2.91%, 2.42%, and 5.88% on ESCI. A product’s own attribute order is therefore sufficient to change its candidate status in these controlled comparisons. Figure 1 illustrates one such case; the population estimates establish its broader scope.

Truncation is not necessary for this sensitivity. Among targets whose seven inputs all fit, VI@20 remains 10.59%, 7.16%, and 14.55% on WANDS and 3.20%, 2.55%, and 5.88% on ESCI (Figure 3). All GTE catalog inputs fit its 8,192-token limit. MiniLM and BGE fitting supports differ from their full populations and from each other; their differences cannot isolate a truncation effect or rank intrinsic encoder sensitivity.

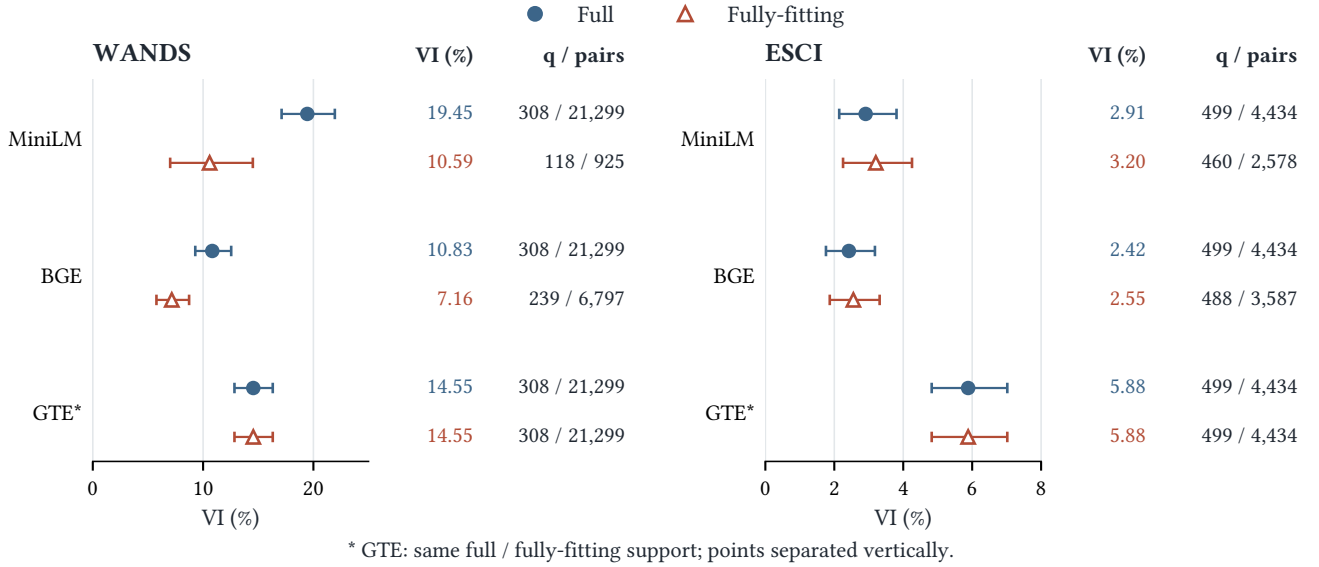


Figure 3: Target-only VI@20 for all eligible targets and fully-fitting targets. Bars show 95% query-cluster intervals; labels report query/pair counts. Dataset panels use different horizontal scales.

Persistent omission also remains substantial. For example, WANDS/BGE catalog-wide retrieval at $K = 100$ yields 57.18% persistent inclusion, 12.30% VI, and 30.51% persistent omission. All three states must be considered: lower instability alone does not establish broader access to relevant products.

Increasing the cutoff improves persistent inclusion but need not reduce VI monotonically. For example, WANDS/MiniLM target-only VI rises from 19.45% at $K = 20$ to 22.45% at $K = 100$, then falls to 10.45% at $K = 1000$ on the same support. Crossings are not uniformly confined to the cutoff neighborhood: at $K = 20$, 12.77% of all WANDS/MiniLM targets cross and have worst rank above 40, and 5.25% cross and exceed 100, using query-macro shares. Corresponding ESCI shares are smaller. The supplementary cutoff and severity analyses distinguish all-target shares from conditional crossing proportions and from source-order losses; rank movement alone does not identify an attention mechanism.

6.2 RQ2: Recall Changes and Relevant-Product Membership

Net Recall changes can understate changes in which relevant products are retrieved. Across the six source-order comparisons at $K = 20$, WANDS membership change is 8.23–10.52% for MiniLM, 4.31–4.77% for BGE, and 5.75–6.36% for GTE. Corresponding net Recall changes range from -2.60 to -0.95 , -0.46 to $+0.14$, and -0.92 to $+0.14$ percentage points. Thus aggregate changes are not uniformly small, but they do not describe the full extent of membership changes. ESCI exhibits smaller changes, which remain nonzero in every encoder/cutoff setting.

Figure 4 separates the two sides of this exchange. For WANDS/BGE under reversal, newly included and newly omitted shares are 2.357% and 2.263%, respectively. Their sum is 4.621%, while

net Recall changes by only $+0.094$ points. Shares use all highest-relevance products within each query before macro aggregation; they are not fractions of the top-20 list.

Cancellation also occurs within individual queries. Across the six WANDS comparisons, exactly equal positive gain and loss counts occur for 52–67 of 308 queries with MiniLM, 67–78 with BGE, and 61–81 with GTE: 16.88–21.75%, 21.75–25.32%, and 19.81–26.30%, respectively. Each affected query has unchanged Recall despite different relevant-product membership. ESCI ranges are 1.40–2.81%, 2.40–3.21%, and 3.01–4.21%. These are ranges over comparisons, not confidence intervals or disjoint query groups. Membership tracking therefore adds information even to a per-query Recall comparison.

6.3 RQ3: Effectiveness and Consistent Inclusion under Controls

Pure canonicalization removes incoming-order dependence by construction. Its effectiveness nevertheless depends on the fixed rule and evaluation setting (Table 1). At $K = 20$, all three rules have lower point estimates than source order for WANDS/MiniLM and for each ESCI encoder; WANDS/BGE and WANDS/GTE include both increases and decreases. A uniform effectiveness cost of invariance is therefore unsupported.

The reference matters. WANDS/BGE lexical ascending yields 63.237% Recall@100: $+0.066$ points relative to source order (63.171%), but -0.383 relative to the raw seven-order mean (63.620%). Stability also does not preserve source-order membership. At $K = 20$, the same rule adds 2.097% and omits 2.129% of relevant products, despite a net change of only -0.032 points. Canonicalization fixes an inclusion decision; that decision still requires evaluation.

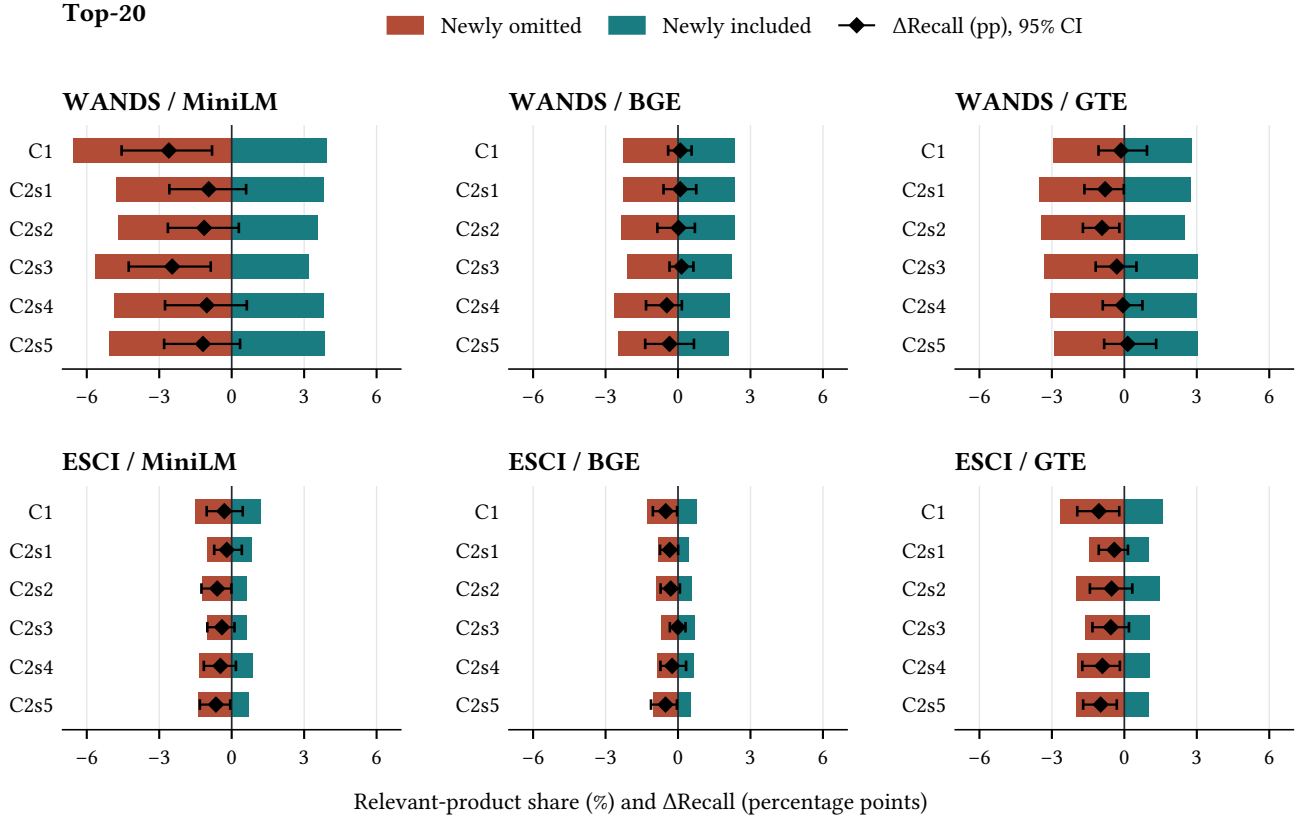


Figure 4: Catalog-wide membership changes relative to source order at $K = 20$. Left/right bars show newly omitted/included shares of H_q ; diamonds show net Recall changes with 95% paired intervals.

Table 1: Raw C0 Recall@20 (%) and paired canonical differences (pp).

Dataset / encoder	C0	Asc.	Desc.	Type
WANDS / MiniLM	30.81	-2.09	-1.25	-1.59
WANDS / BGE	36.10	-0.03	+0.17	+0.38
WANDS / GTE	35.90	-0.16	-0.90	+0.08
ESCI / MiniLM	67.95	-0.46	-0.43	-0.47
ESCI / BGE	74.10	-0.26	-0.51	-0.29
ESCI / GTE	72.45	-0.84	-0.80	-0.77

Asc., Desc., and Type are the three fixed pure rules. Each has structural VI=0. Full paired 95% intervals are supplied in Table 3 and the artifact; no held-out rule is selected.

Aggregation outcomes are likewise setting-dependent (Table 2). Set-Mean has lower Recall@100 than source order in both BGE settings and on ESCI/MiniLM, but increases it on WANDS/MiniLM. Full-record aggregation with $M = 2$ improves point estimates for BGE on both datasets and for ESCI/MiniLM, while reducing them for WANDS/MiniLM. These observations do not identify a single encoding-unit effect.

More views do not produce monotonic Recall improvements. For WANDS/BGE, centroid Recall@100 is 63.956%, 64.394%, and 64.156%

at $M = 2, 4, 7$; corresponding ESCI values are 91.066%, 91.014%, and 91.013%. The BGE $M = 7$ family produces a mean of 6.995 distinct text views per WANDS product but only 3.402 per ESCI product. Centroids retain one retrieval vector, whereas multi-vector retrieval stores M vectors. View diversity and encoding/storage costs therefore accompany effectiveness comparisons; these results do not establish that diversity causes the dataset differences.

Hybrid increases seven-schedule mean Recall@100 from 56.613% to 65.548% for WANDS/MiniLM, 63.620% to 67.773% for WANDS/BGE, 86.193% to 89.567% for ESCI/MiniLM, and 91.041% to 91.610% for ESCI/BGE. Paired changes are +8.93 [7.21, 10.82], +4.15 [2.55, 5.82], +3.37 [2.18, 4.69], and +0.57 [-0.21, 1.37] percentage points, respectively. VI@100 decreases by 10.81, 5.26, and 0.85 points in the first three settings, while ESCI/BGE increases by 0.025 points with an interval of [-0.409, 0.572]. The last setting provides descriptive co-occurrence of higher mean Recall and VI, but neither interval excludes zero. It does not establish a general effectiveness-instability trade-off.

Persistent inclusion also increases at $K = 100$, by 15.01, 7.08, 3.79, and 0.52 points in the same order. These joint outcomes differ from a single source-order Recall comparison. Deterministic controls remove incoming-order variation but still retrieve different products; hybrid retains order dependence while often reducing its measured

Table 2: Joint strategy evaluation: mean Recall@100 / VI@100 (%).

Method	WANDS		ESCI	
	MiniLM	BGE	MiniLM	BGE
Raw (seven schedules)	56.61 / 21.89	63.62 / 12.30	86.19 / 1.41	91.04 / 0.72
Canonical ascending	56.05 / 0.00	63.24 / 0.00	86.41 / 0.00	91.05 / 0.00
Canonical descending	56.16 / 0.00	63.61 / 0.00	86.38 / 0.00	91.01 / 0.00
Canonical type/style	57.19 / 0.00	63.56 / 0.00	86.36 / 0.00	91.09 / 0.00
Set-Mean	58.81 / 0.00	61.82 / 0.00	84.50 / 0.00	88.95 / 0.00
Centroid, $M = 2$	56.44 / 0.00	63.96 / 0.00	86.33 / 0.00	91.07 / 0.00
Multi-vector max, $M = 2$	57.04 / 0.00	64.00 / 0.00	86.35 / 0.00	91.09 / 0.00
BM25	63.23 / 0.00	63.23 / 0.00	82.83 / 0.00	82.83 / 0.00
Hybrid (seven schedules)	65.55 / 11.08	67.77 / 7.05	89.57 / 0.56	91.61 / 0.74

Raw and hybrid average all seven schedules. Canonical, Set-Mean, canonical-seeded centroid/max and BM25 use a fixed input-order independent index: their per-schedule Recall is identical and structural VI is zero. All three canonical rules are retained. WANDS: 308 queries, 21,299 Exact pairs; ESCI: 499 queries, 4,434 E pairs. Source-order Recall, persistent states, paired intervals, membership counts and cost are reported in the artifact tables.

extent. The evidence supports evaluating effectiveness, consistent inclusion, product membership, and cost together.

A Additional Evaluation Details

Lexical canonicalization sorts each complete attribute entry by the tuple `(entry.casefold(), entry)`, ascending or descending as specified. The type/style rule extracts the text before the first colon and lowercases it; without a colon, it uses the first 40 characters instead. Entries whose extracted text contains `product type`, `type`, or `style` are placed first. Each group uses ascending lexical order with the same tuple. Duplicate occurrences remain. Unkeyed entries receive no invented field names. Historical template-based canonicalization changes text framing and is excluded from the pure-sorting comparisons.

For each shuffle seed and product identifier, the first eight bytes of SHA256 applied to `seed:product_id`, interpreted as a big-endian integer, seed Python’s random shuffle. Raw schedules start from source order; full-record aggregation starts from the canonical ascending list and uses its separate seed sequence. We retain repeated views, use the requested M with equal weight, and record distinct texts and vectors separately. In ESCI, the complete bullet-point field is a single entry; with brand and color, the maximum number of distinct entry permutations is six.

MiniLM, BGE, and GTE use pinned revision prefixes 1110a243fdf4, a5beb1e3e68b, and e7f32e3c00f9, respectively. Full repository identifiers and revisions are in the frozen protocol. CUDA encoding uses FP16 model weights and FP32 pooling and normalization. Historical query embeddings and authoritative saved ranks are reused. Independent floating-point encoder executions are not assumed bitwise identical.

Fitting eligibility uses untruncated token counts including special tokens for every one of the seven inputs. Table 5 gives the restricted supports. Retrieval keeps the complete original catalog. The illustrative WANDS query *teal chair*, product 24318, has ranks 11 and 939 under C0 and C2s5 with 42,993 competitors fixed. Its seven inputs each contain 1,059 tokens under GTE’s 8,192-token limit. This post-hoc example does not estimate typical rank displacement.

The historical WANDS split orders query identifiers by SHA256 of `phase3-dev:query_id`, retaining 96 development and 384 held-out identifiers. ESCI selects 500 query identifiers by SHA256 of `20260904:query_id` from the US small-version test split. Highest-label-free queries are excluded from Recall and inclusion summaries. The historical development-selected appearance rule is distinct from the fixed type/style rule reported here. BGE’s $M = 4, 7$ results retain their existing fixed-budget comparison role; a held-out maximum is not presented as a tuned deployment choice.

All completion-package intervals, including BGE view-budget contrasts, use 10,000 paired query-cluster draws (seed 2026091701). Historical seed-20260907 intervals are not used in the common main or supplementary strategy tables. Existing cost measurements use an Intel i7-13700H CPU and RTX 4060 Laptop GPU with 8 GB memory. Exact CPU scoring uses four threads, the first 50 queries, one warm-up, and five measured repetitions. Reported array storage excludes vocabulary, metadata, and approximate-search overhead. Cached encoding time is not treated as a matched cold-build benchmark.

Target-only severity uses best and worst ranks over the same seven schedules. A crossing satisfies $\text{best} \leq K < \text{worst}$. All-target

severe-crossing shares divide by every product in H_q before query averaging. Conditional query-macro shares first divide by a query’s crossing count and then average only queries with crossings. Pair-micro conditional fractions weight crossing pairs equally. The artifact also reports the ratio of query-normalized severe and crossing mass, which is a different weighting. Source-order-directed losses additionally require C0 inclusion and omission in at least one alternative schedule. ECDFs and rank-range distributions retain both full and fully-fitting supports; the complete competitor catalog remains.

Table 3: Complete pure-canonical results at $K = 20$. Recall and shares are percentages; differences are percentage points.

Dataset / encoder	Rule	Recall	Δ C0 [95% CI]	Δ raw mean [95% CI]	Gain / loss
WANDS / MiniLM	Ascending	28.72	-2.09 [-4.03, -0.22]	-0.75 [-2.05, +0.48]	3.72 / 5.80
WANDS / MiniLM	Descending	29.56	-1.25 [-2.69, +0.06]	+0.09 [-1.12, +1.30]	3.45 / 4.70
WANDS / MiniLM	Type/style	29.22	-1.59 [-3.49, +0.22]	-0.25 [-1.68, +1.09]	3.81 / 5.40
WANDS / BGE	Ascending	36.07	-0.03 [-0.53, +0.45]	+0.03 [-0.46, +0.53]	2.10 / 2.13
WANDS / BGE	Descending	36.28	+0.17 [-0.72, +1.16]	+0.24 [-0.50, +1.09]	2.43 / 2.26
WANDS / BGE	Type/style	36.48	+0.38 [-0.08, +0.89]	+0.45 [+0.00, +0.93]	2.37 / 1.99
WANDS / GTE	Ascending	35.75	-0.16 [-1.08, +0.87]	+0.14 [-0.47, +0.84]	3.18 / 3.34
WANDS / GTE	Descending	35.00	-0.90 [-1.70, -0.15]	-0.60 [-1.20, -0.02]	2.63 / 3.53
WANDS / GTE	Type/style	35.98	+0.08 [-0.56, +0.68]	+0.38 [-0.18, +0.92]	2.84 / 2.76
ESCI / MiniLM	Ascending	67.50	-0.46 [-0.98, +0.05]	-0.08 [-0.49, +0.34]	0.52 / 0.98
ESCI / MiniLM	Descending	67.52	-0.43 [-1.13, +0.30]	-0.06 [-0.42, +0.32]	0.86 / 1.29
ESCI / MiniLM	Type/style	67.48	-0.47 [-1.00, +0.03]	-0.10 [-0.51, +0.33]	0.53 / 1.00
ESCI / BGE	Ascending	73.84	-0.26 [-0.65, +0.09]	+0.01 [-0.30, +0.33]	0.54 / 0.80
ESCI / BGE	Descending	73.60	-0.51 [-0.95, -0.12]	-0.23 [-0.59, +0.09]	0.54 / 1.05
ESCI / BGE	Type/style	73.82	-0.29 [-0.68, +0.08]	-0.01 [-0.32, +0.30]	0.53 / 0.82
ESCI / GTE	Ascending	71.61	-0.84 [-1.70, -0.03]	-0.21 [-0.79, +0.38]	1.12 / 1.96
ESCI / GTE	Descending	71.64	-0.80 [-1.60, -0.06]	-0.17 [-0.74, +0.37]	1.06 / 1.87
ESCI / GTE	Type/style	71.68	-0.77 [-1.62, +0.04]	-0.13 [-0.71, +0.44]	1.19 / 1.96

Query-macro estimates, 10,000 paired query-cluster draws, seed 2026091701. Intervals are descriptive and uncorrected; zero in an interval is not equivalence. Gains/losses compare each actual canonical list to raw C0; no membership list is assigned to the raw seven-order mean.

Table 4: Complete pure-canonical results at $K = 100$. Recall and shares are percentages; differences are percentage points.

Dataset / encoder	Rule	Recall	Δ C0 [95% CI]	Δ raw mean [95% CI]	Gain / loss
WANDS / MiniLM	Ascending	56.05	-1.83 [-3.17, -0.64]	-0.56 [-1.52, +0.31]	3.52 / 5.35
WANDS / MiniLM	Descending	56.16	-1.72 [-2.97, -0.58]	-0.45 [-1.36, +0.43]	3.93 / 5.65
WANDS / MiniLM	Type/style	57.19	-0.70 [-2.12, +0.68]	+0.57 [-0.59, +1.77]	4.24 / 4.94
WANDS / BGE	Ascending	63.24	+0.07 [-1.47, +1.65]	-0.38 [-1.40, +0.56]	3.20 / 3.13
WANDS / BGE	Descending	63.61	+0.44 [-0.39, +1.47]	-0.01 [-0.61, +0.58]	2.63 / 2.19
WANDS / BGE	Type/style	63.56	+0.39 [-0.97, +1.86]	-0.06 [-1.00, +0.88]	3.26 / 2.86
WANDS / GTE	Ascending	62.63	-0.22 [-1.27, +0.76]	+0.06 [-0.44, +0.61]	2.83 / 3.05
WANDS / GTE	Descending	61.89	-0.96 [-2.29, +0.23]	-0.68 [-1.67, +0.16]	2.82 / 3.78
WANDS / GTE	Type/style	62.63	-0.22 [-1.20, +0.63]	+0.07 [-0.44, +0.56]	2.71 / 2.93
ESCI / MiniLM	Ascending	86.41	+0.28 [-0.05, +0.76]	+0.22 [+0.00, +0.47]	0.38 / 0.11
ESCI / MiniLM	Descending	86.38	+0.25 [-0.22, +0.81]	+0.19 [-0.04, +0.46]	0.57 / 0.33
ESCI / MiniLM	Type/style	86.36	+0.22 [-0.09, +0.70]	+0.16 [-0.05, +0.40]	0.36 / 0.14
ESCI / BGE	Ascending	91.05	+0.07 [-0.04, +0.18]	+0.00 [-0.10, +0.11]	0.18 / 0.11
ESCI / BGE	Descending	91.01	+0.04 [-0.13, +0.19]	-0.03 [-0.13, +0.08]	0.18 / 0.15
ESCI / BGE	Type/style	91.09	+0.11 [-0.01, +0.23]	+0.04 [-0.05, +0.15]	0.22 / 0.11
ESCI / GTE	Ascending	88.80	-0.32 [-0.62, -0.06]	+0.03 [-0.34, +0.44]	0.16 / 0.48
ESCI / GTE	Descending	88.71	-0.42 [-1.18, +0.27]	-0.06 [-0.47, +0.33]	0.41 / 0.83
ESCI / GTE	Type/style	88.83	-0.30 [-0.60, -0.04]	+0.06 [-0.31, +0.47]	0.19 / 0.48

Query-macro estimates, 10,000 paired query-cluster draws, seed 2026091701. Intervals are descriptive and uncorrected; zero in an interval is not equivalence. Gains/losses compare each actual canonical list to raw C0; no membership list is assigned to the raw seven-order mean.

Table 5: Fully-fitting target support and target-only VI@20.

Dataset	Encoder	Queries	Pairs	VI (%)
WANDS	MiniLM	118	925	10.59
WANDS	BGE	239	6,797	7.16
WANDS	GTE	308	21,299	14.55
ESCI	MiniLM	460	2,578	3.20
ESCI	BGE	488	3,587	2.55
ESCI	GTE	499	4,434	5.88

Table 6: Exact within-query cancellation across six attribute-order comparisons at Top-20.

Dataset	Encoder	Eligible q	Queries (%)
WANDS	MiniLM	308	16.88–21.75
WANDS	BGE	308	21.75–25.32
WANDS	GTE	308	19.81–26.30
ESCI	MiniLM	499	1.40–2.81
ESCI	BGE	499	2.40–3.21
ESCI	GTE	499	3.01–4.21

Newly included = newly omitted > 0. Six-comparison min–max ranges, not confidence intervals; denominators include all eligible queries. Counts are not summed across comparisons.

References

- [1] Yan Chen, Shujian Liu, Zheng Liu, Weiyi Sun, Linas Baltrunas, and Benjamin Schroeder. 2022. WANDS: Dataset for Product Search Relevance Assessment. In *European Conference on Information Retrieval*.
- [2] Chandan K. Reddy, Lluís Màrquez, Fran Valero, Nikhil Rao, Hugo Zaragoza, Sambaran Bandyopadhyay, Arnab Biswas, Anlu Xing, and Karthik Subbian. 2022. Shopping Queries Dataset: A Large-Scale ESCI Benchmark for Improving Product Search. *arXiv preprint arXiv:2206.06588* (2022).